

Natural Killer Cell and Glucocorticoid Deficiency with DNA Repair Defect (MCM4 Deficiency)

MONDO:0012383 · OMIM #609981 · Immunodeficiency-54 (IMD54)

Comprehensive disease-characteristics report. Evidence is predominantly disease-level (aggregated from a small number of related patients and from model organisms), not EHR-derived. Human cohorts are very small (~2 index families, <30 patients), so many quantitative fields rest on case-series and cross-species inference; these limits are flagged throughout.

Summary (Answer to the Research Question)

Natural Killer Cell and Glucocorticoid Deficiency with DNA Repair Defect is an ultra-rare **autosomal recessive** multisystem disorder caused by **hypomorphic biallelic mutations in MCM4** (minichromosome maintenance complex component 4), a subunit of the **MCM2-7 replicative DNA helicase**. The founder allele **c.71-1insG (p.Pro24ArgfsX4)** was discovered in a genetically isolated **Irish** population. Because MCM4 licenses DNA replication origins, the hypomorphic defect depletes "dormant" backup origins and produces replication stress and **genomic instability** that manifest most in cells with high proliferative demand. The clinical triad is (1) **primary adrenal / glucocorticoid insufficiency**, (2) **pre- and postnatal growth retardation / short stature**, and (3) **a selective natural killer (NK) cell deficiency** (loss of the mature CD56^{dim} subset), accompanied by a laboratory signature of increased chromosomal breakage and an inferred elevated cancer risk. Two landmark 2012 papers (Hughes et al.,

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Gineau

et

al.,

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independently established the gene, mechanism, and phenotype.

1. Disease Information

Overview. A genetic syndrome combining ACTH-resistant primary adrenal failure, growth failure, and an NK-cell immunodeficiency with an underlying DNA-replication/repair defect. It is a mechanistically distinct variant within the **familial glucocorticoid deficiency (FGD)** family of disorders and within the **NK cell deficiency (NKD)** family of inborn errors of immunity.

Key identifiers. - **MONDO:** MONDO:0012383 - **OMIM:** #609981 (Natural killer cell and glucocorticoid deficiency with DNA repair defect / IMD54) - **Orphanet:** Rare disorder (grouped with familial glucocorticoid deficiency / primary NK-cell immunodeficiency; ORPHA classification "rare") - **Gene / HGNC:** MCM4, HGNC:6947; NCBI Gene 4173; UniProt P33991; cytoband 8q11.21 - **MeSH concepts:** "Minichromosome Maintenance Complex Component 4"; "Killer Cells, Natural"; "Adrenal Insufficiency" - **ICD-10:** best mapped to E27.4 (other/unspecified adrenocortical insufficiency) + D84.9 (immunodeficiency, unspecified); **ICD-11:** ~4A00 (primary immunodeficiencies) + 5A74 (adrenocortical insufficiency). No dedicated ICD code.

Synonyms / alternative names. MCM4 deficiency; Immunodeficiency 54 (IMD54); Natural killer cell deficiency with adrenal insufficiency and growth retardation; Familial glucocorticoid deficiency with NK cell deficiency; NKD associated with DNA repair/replication defect.

Information source. Aggregated disease-level (OMIM/Orphanet + primary case series and model-organism studies), not individual EHR data.

2. Etiology

Primary cause — genetic. Biallelic hypomorphic loss-of-function variants in **MCM4**. The reported human allele **c.71-1insG** produces a frameshift (p.Pro24ArgfsX4) predicted to truncate the protein; the allele is **hypomorphic** because downstream re-initiation methionine codons regenerate a shorter, partially functional ~85 kDa isoform while the normal 96 kDa isoform

is

absent

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Complete MCM4 loss is embryonic-lethal (mouse *Mcm4*^{-/-} is preimplantation-lethal; P 17143284), so only hypomorphic alleles are compatible with live birth.

Genetic risk factors. The causal variant is the sole determinant. **Consanguinity / endogamy** (Irish isolate) and **carrier status of both parents** are the operative risk factors. No common susceptibility loci or GWAS signals apply (monogenic disease).

Environmental risk factors. None initiate the disease. However, **DNA-damaging exposures** (e.g., aphidicolin in vitro; by extension ionizing radiation, replication-stress-inducing agents) markedly worsen the cellular phenotype — patient and *Mcm4*^{Chaos3} fibroblasts are hypersensitive to aphidicolin-induced chromosome breakage (P 17143284).

P 22354167 Viral exposures (herpesviruses) act as **triggers of clinical morbidity** given the NK defect.

Protective factors. No established genetic modifiers or protective alleles in humans. Mechanistically, anything that lowers replication stress (adequate nucleotide pools) or that restores WT MCM4 (demonstrated in vitro: WT MCM4 expression rescues genomic instability; P 22354167) is protective. No dietary/lifestyle protective factors are known.

Gene–environment interaction. The core interaction is **genotype (hypomorphic MCM4) × replication stress**: environmental/chemical replication inhibitors convert a tolerable origin-licensing deficit into overt fork collapse and chromosome breaks. Mouse genetics show epistasis with DNA-damage-response genes: *Mcm4*^{Chaos3} combined with *Atm*, *p21/Cdkn1a* or *Chk2* deficiency reduces tumor latency (P 23975433), and with *Fancc* loss increases genome instability (P 24589582).

3. Phenotypes

Reported across the two human cohorts (Irish FGD family; French/Irish related patients) and OMIM. Frequencies are qualitative given <30 patients.

Phenotype	Type	HPO term	Onset	Severity	Frequency
Glucocorticoid (adrenal) insufficiency, ACTH-resistant	Lab/clinical	HP:0008207 (Primary adrenal insufficiency) / HP:0000833	Infancy–childhood	Moderate–severe, treatable	Very frequent (hallmark)
Hypocortisolemia with elevated ACTH	Lab	HP:0008163 (Decreased circulating cortisol); HP:0003189-like elevated ACTH	Childhood	—	Very frequent
Pre- and postnatal growth retardation / short stature	Physical	HP:0004322 (Short stature); HP:0001511 (IUGR)	Congenital/childhood	Moderate–severe	Very frequent (hallmark)
NK cell deficiency (selective CD56 ⁺ dim loss)	Lab/immune	HP:0040218 (Reduced NK cell count) / HP:0012176 abnormal NK physiology	Congenital	Severe, persistent	Very frequent (hallmark)
Susceptibility to viral (herpesvirus) infection	Clinical	HP:0004429 (Recurrent viral infections); HP:0032152 (Herpesvirus infection)	Childhood	Variable, can be severe/fatal	Frequent
Increased chromosomal breakage / genomic instability	Lab	HP:0040012 (Chromosome breakage)	Congenital (cellular)	—	Very frequent (in vitro)
Microcephaly / small head (reported in some)	Physical	HP:0000252	Congenital	Mild–moderate	Variable
Skin/pigmentary changes	Physical	HP:0000953 (Hyperpigmentation)	Childhood	Mild	Variable

Phenotype	Type	HPO term	Onset	Severity	Frequency
(hyperpigmentation from high ACTH)					
Increased neoplasia risk (inferred)	Clinical	HP:0002664 (Neoplasm)	Adult (inferred)	Potentially severe	Unquantified in humans

Quality-of-life impact. Chronic lifelong disease: dependence on daily steroid replacement (with risk of adrenal crisis under stress), short stature affecting growth/psychosocial wellbeing, and recurrent infection burden. No formal EQ-5D/SF-36 data exist for this ultra-rare disorder.

4. Genetic / Molecular Information

Causal gene. **MCM4** (HGNC:6947; OMIM 602638; 8q11.21). *Encodes a subunit of the heterohexameric MCM2-7 complex — the replication-licensing factor and the motor core of the CMG (Cdc45–MCM–GINS) replicative helicase** (P 30369561).

Pathogenic variant (human). - **c.71-1insG; p.Pro24ArgfsX4** — splice-acceptor/insertion frameshift, **germline**, homozygous. Classification: **pathogenic** (segregates recessively, functional rescue by WT). - **Variant class:** frameshift/splice-region; **functional consequence:** hypomorphic **loss of function** (major isoform absent, minor 85 kDa isoform preserved via downstream translation re-initiation) (P 22354170).

(P 22354167).
- **Allele frequency:** ultra-rare / effectively private founder allele; enriched in one Irish isolate; not a common gnomAD allele. - **Somatic vs germline:** germline (constitutional).

Modifier genes. None validated in humans. Mouse data implicate the **ATM–CHK2–p21/TP53** DNA-damage-response axis and the **Fanconi anemia (FANCC)** pathway as genetic modifiers of instability/tumorigenesis (P 23975433).

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and **miR-34** / **TRP53** as post-transcriptional modifiers of MCM2-7 levels

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Epigenetic information. No disease-specific methylation/histone signature reported. TP53-dependent miR-34 upregulation reduces MCM2-7 mRNA under replication stress

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Chromosomal abnormalities. No constitutional karyotype abnormality; the defect manifests as **acquired chromosomal instability** — increased breaks/gaps, micronuclei, and (in mouse models) amplifications/deletions including activating *Notch1* deletions in Sdl leukemias

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5. Environmental Information

- **Environmental factors:** DNA-damaging / replication-stress agents (aphidicolin experimentally; ionizing radiation and genotoxins by extension) aggravate the cellular phenotype. Not disease-initiating.
- **Lifestyle factors:** none causal; standard adrenal-insufficiency precautions (stress-dose steroids during illness) are relevant to outcomes.
- **Infectious agents:** herpesviruses — **cytomegalovirus (CMV; NCBI:txid10359)**, **varicella zoster virus (VZV; txid10335)**, **Epstein–Barr virus (EBV; txid10376)** — are the principal clinically important pathogens, acting as triggers rather than causes

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6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic hypomorphic MCM4 (c.71-1insG)** → loss of the full-length 96 kDa MCM4 isoform, retaining only a partly functional 85 kDa isoform (*demonstrated*,

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2. → **Destabilized MCM2-7 helicase / reduced functional MCM loading** onto origins (*demonstrated* in yeast/mouse orthologs; MCM4:MCM6 interaction disrupted in Chaos3,

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3. → **Depletion of licensed "dormant" replication origins** (backup origins for stalled forks) (*demonstrated* in MCM-hypomorphic mice,

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4. → **Replication stress: stalled/collapsed forks, incomplete replication** (*demonstrated* in patient fibroblasts and MEFs; aphidicolin hypersensitivity,

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5. → **Genomic instability (chromosome breaks, micronuclei, copy-number changes)** (*demonstrated*; rescued by WT MCM4,

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6. Branch point — the ubiquitous defect produces **tissue-heterogeneous** outcomes concentrated in high-turnover / low-tolerance compartments:
7. **6a. Immune branch:** Generation of terminally mature **CD56^{dim} NK cells requires extra MCM4-dependent divisions**; CD56^{bright} cells proliferate poorly and are apoptosis-prone under replication stress → **selective loss of CD56^{dim} NK subset** → **impaired antiviral NK immunity** → **herpesvirus susceptibility** (*demonstrated*,

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8. **6b. Adrenal branch:** Impaired proliferation/maintenance of **steroidogenic zona fasciculata cells**, which are replaced by **non-steroidogenic GATA4⁺/Gli1⁺ cells** → reduced cortisol output → **ACTH-resistant glucocorticoid deficiency** (*demonstrated in Mcm4-depleted mice*,

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9. **6c. Growth branch:** Ubiquitous replication limitation across proliferating tissues → **pre-/postnatal growth retardation, short stature** (*inferred from ubiquitous impact*,

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10. **6d. Oncogenic branch:** Persistent instability + reduced dormant-origin buffering → **elevated neoplastic risk** (*demonstrated in mouse; inferred in humans*,

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11. Downstream modulation → **TP53/ATM–CHK2 activation, p21, miR-34-mediated MCM2-7 downregulation, senescence/apoptosis** shape which cells die vs transform (*demonstrated in models*,

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Category detail

- **Molecular pathways:** DNA replication initiation/licensing (KEGG hsa03030 DNA replication; Reactome R-HSA-69306 "DNA Replication", R-HSA-68867 "Assembly of the pre-replicative complex"); ATR/ATM DNA-damage checkpoint signaling; TP53 pathway.
 - **Cellular processes:** replication-origin licensing, S-phase progression, cell-cycle checkpoint, replication-stress response, apoptosis, senescence, impaired proliferation/ differentiation.
- GO:** GO:0006270 (DNA replication initiation), GO:0000727 (double-strand break repair via break-induced replication), GO:0031570 (DNA integrity checkpoint), GO:0006281 (DNA repair), GO:0006260 (DNA replication), GO:0008283 (cell population proliferation).

- **Protein dysfunction:** destabilized MCM4 → weakened MCM2-7 hexamer assembly and helicase loading (loss of function, not aggregation).
 - **Immune involvement:** primary immunodeficiency (NK-cell); no autoimmunity signature.
 - **Tissue damage:** replication-stress-driven cell attrition and genome instability rather than ischemia/fibrosis.
 - **Cell types (CL):** NK cell CL:0000623 (specifically CD56^{dim} mature NK, CD56^{bright} NK precursor); adrenal cortical/steroidogenic cell CL:0000541-like (zona fasciculata cell); fibroblast CL:0000057. **Subcellular (GO CC):** nucleus GO:0005634; MCM complex GO:0042555; CMG complex GO:0071162; replication fork GO:0005657.
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7. Anatomical Structures Affected

- **Organ level (primary): Adrenal gland** (UBERON:0002369; specifically adrenal cortex UBERON:0001235, zona fasciculata UBERON:0001233) — endocrine system; **immune / hematopoietic system** (NK cells; bone marrow UBERON:0002371, blood UBERON:0000178, spleen/lymphoid tissue).
 - **Secondary / systemic:** skeletal growth (short stature — musculoskeletal system); potential multi-organ neoplasia risk; skin (hyperpigmentation from ACTH excess, UBERON:0002097).
 - **Body systems:** endocrine, immune, and (growth) musculoskeletal.
 - **Tissue / cell level:** adrenal steroidogenic epithelial/endocrine cells; peripheral-blood NK lymphocytes (CD56^{dim} CL); fibroblasts show the cellular defect.
 - **Subcellular:** nucleus / chromatin, replication fork, MCM2-7 and CMG helicase complexes.
 - **Localization / laterality:** bilateral (adrenal, systemic/generalized); not lateralized.
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8. Temporal Development

- **Onset:** congenital cellular defect; clinical onset in **infancy–childhood**. Growth retardation is **pre- and postnatal**; adrenal insufficiency and infections present in early childhood. **Insidious/chronic** onset.

- **Progression:** chronic, **lifelong**, generally **stable** with treatment of the endocrine and infectious components; the genomic-instability/cancer risk is **progressive/age-dependent** (inferred). NK deficiency is persistent and non-remitting.
- **Disease course:** non-episodic; adrenal crises can occur acutely under physiologic stress if steroid replacement is inadequate.
- **Critical periods / intervention windows:** neonatal–early childhood recognition of adrenal insufficiency is life-saving (prevent hypoglycemia/adrenal crisis); early NK-defect recognition guides antiviral vigilance.

9. Inheritance and Population

- **Epidemiology:** ultra-rare; <30 **reported patients** worldwide, clustered in an Irish isolate plus related patients. No reliable prevalence/incidence rates (Orphanet: "rare", point prevalence unknown, likely <1/1,000,000).
- **Inheritance:** **autosomal recessive** (OMIM #609981).
- **Penetrance:** high/complete for the core biochemical/immunologic phenotype in homozygotes; expressivity of viral disease and growth is **variable**.
- **Anticipation / mosaicism:** not applicable / not reported.
- **Founder effect: yes** — c.71-1insG enriched in a genetically isolated Irish (Traveller) community; homozygosity in consanguineous/endogamous pedigrees
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- **Consanguinity:** contributory (related patients mapped by homozygosity/linkage).
- **Carrier frequency:** elevated within the founder population; low/negligible in the general population.
- **Demographics:** described in individuals of **Irish** ancestry; **no sex predilection** (autosomal); pediatric age at diagnosis.

10. Diagnostics

Clinical / laboratory tests. - **Endocrine:** low serum/plasma **cortisol** (LOINC 2143-6) with **elevated ACTH** (LOINC 2141-0); normal mineralocorticoid/renin (glucocorticoid-selective) — pattern of ACTH-resistant primary adrenal insufficiency. - **Immunophenotyping (flow cytometry):** reduced/absent **CD56^{dim}** NK subset with relatively preserved **CD56^{bright}** cells; overall low NK cytotoxicity. This is the discriminating immunologic test. - **Cytogenetic / DNA-repair assay:** increased **chromosomal breakage**, spontaneous and aphidicolin-induced, in cultured fibroblasts/lymphocytes (functional signature). - **Auxology / imaging:** documentation of growth failure; adrenal imaging may show small/ abnormal adrenals (mechanistic correlate of steroidogenic-cell loss).

Genetic testing. Definitive diagnosis by demonstrating **biallelic MCM4 variants**: single-gene MCM4 sequencing (targeted for the founder allele in Irish patients), **inborn- errors-of-immunity / adrenal-insufficiency gene panels**, **whole-exome (WES)** or **whole-genome (WGS)** sequencing (both original discoveries used targeted exome / linkage). CMA/karyotype/FISH are not diagnostic (no constitutional structural lesion). Cascade testing of relatives for the familial variant.

Clinical criteria / differential diagnosis. No formal consensus criteria; diagnosis is the combination of the triad + molecular confirmation. **Differentials:** - Other **familial glucocorticoid deficiency** genes: **MC2R** (FGD1), **MRAP** (FGD2), **NNT**, **MCM4** (this disease), **TXNRD2**, **STAR**, **AAAS** (triple-A/Allgrove) — distinguished by additional features and gene testing

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- Other **NK cell deficiencies / helicaseopathies:** **GINS1**, **MCM10**, **GINS4**, **POLE1**, **RTEL1** — overlapping NK defect but different systemic features

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- **Chromosomal instability / DNA-repair syndromes** (Fanconi anemia, Nijmegen breakage) — distinguished by their own gene panels and phenotypes.

Screening. No population newborn screen; **carrier screening** feasible within the founder community; cascade and prenatal/preimplantation testing available once the familial variant is known.

11. Outcome / Prognosis

- **Survival / mortality:** no formal survival statistics (too few patients). Prognosis is **largely determined by two treatable/monitorable threats:** (i) **adrenal crisis** (preventable with steroid replacement) and (ii) **severe viral (herpesvirus) infection**, which in NKD can be **fatal**

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An inferred **lifetime cancer risk** from genomic instability adds long-term mortality risk (mouse models: >80% mammary tumors; human risk unquantified —

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- **Morbidity / function:** chronic burden of steroid dependence, short stature, and infection susceptibility; no standardized disability/QoL metrics available.
- **Complications:** adrenal crisis/hypoglycemia; recurrent or severe herpesviral disease (CMV/EBV/VZV); potential malignancy.
- **Recovery potential:** endocrine and infection risks are manageable; the underlying replication defect is not reversible without correcting MCM4.
- **Prognostic factors:** timeliness of adrenal-insufficiency treatment, severity/frequency of viral infections, and (theoretically) tumor surveillance. **Prognostic biomarker:** degree of NK CD56^{dim} depletion and chromosomal-breakage burden (mechanistic, not validated clinically).

12. Treatment

No disease-specific approved therapy exists; management is organ-directed and supportive.

- **Pharmacotherapy — endocrine replacement (mainstay): hydrocortisone** (glucocorticoid replacement; NCIT C312 / CHEBI:17650 cortisol) titrated to need, with **stress dosing** during illness/surgery and patient/family education on adrenal-crisis prevention (parenteral hydrocortisone for crisis). Mineralocorticoids generally **not** required (glucocorticoid- selective defect). **NCIT:** Hydrocortisone Therapy.
- **Antiviral / infection management:** vigilant surveillance and prompt treatment of herpesviral infections (e.g., **acyclovir/valacyclovir** for HSV/VZV; **ganciclovir/valganciclovir** for CMV); consideration of prophylaxis in individual patients. IVIG is not a standard indication unless additional humoral defects arise. (Extrapolated from NKD management principles,

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- **Growth:** monitor growth; endocrinology-guided management (GH therapy has no established role specific to this disorder).
- **Advanced therapeutics: Hematopoietic stem cell transplantation (HSCT)** can correct the hematopoietic/NK compartment in NKD and has been used across DNA-repair-associated immunodeficiencies, but conditioning carries **increased genotoxic/regimen-related toxicity risk** in DNA-repair-defective patients; **reduced-intensity conditioning** is favored (analogy to POLD1 deficiency HSCT,

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HSCT does **not** correct the adrenal or growth (non-hematopoietic) phenotype. No approved gene therapy; MCM4 gene correction rescues the cellular phenotype in vitro

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— proof-of-concept only.

- **Supportive care:** medical-alert identification, emergency steroid kit, immunization per IEI guidance (caution with live-attenuated vaccines given NK/viral susceptibility), tumor surveillance in long-term follow-up (inferred prudence).
- **Experimental / clinical trials:** none specific to MCM4 deficiency (ClinicalTrials.gov).

- **Pharmacogenomics / personalized medicine:** genotype-guided family counseling; caution with genotoxic chemotherapy/radiotherapy given intrinsic replication-stress sensitivity.

13. Prevention

- **Primary prevention:** not preventable once biallelic MCM4 present; **genetic counseling** and **carrier screening** in at-risk (founder/consanguineous) families reduce recurrence; **prenatal diagnosis** and **preimplantation genetic testing** available for known familial variant.
- **Secondary prevention:** early recognition of adrenal insufficiency (prevent crisis/hypoglycemia) and early NK-defect identification (guide antiviral vigilance); no population newborn screen exists.
- **Tertiary prevention:** stress-dose steroid protocols, prompt antiviral therapy, avoidance of unnecessary genotoxic exposures, and long-term malignancy surveillance.
- **Immunization:** routine inactivated vaccines; **caution with live-attenuated vaccines** given NK/viral susceptibility.
- **Counseling:** autosomal-recessive recurrence risk 25% per pregnancy for carrier couples; cascade testing of relatives (NSGC/ACMG frameworks).

14. Other Species / Natural Disease

- **Taxonomy:** experimentally modeled in **Mus musculus** (NCBI:txid10090) and **Saccharomyces cerevisiae** (txid4932); no reported spontaneous companion-animal/wildlife disease equivalent (OMIA: no established natural MCM4 disorder entry).
- **Orthologous genes:** mouse **Mcm4** (NCBI Gene 17217; MGI:1298225); yeast **CDC54/MCM4**. MCM4 and the MCM2-7 complex are **highly evolutionarily conserved** across all eukaryotes
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so disease mechanisms translate strongly across species.
- **Comparative biology:** the *Mcm4*^{Chaos3} (F345I) mouse reproduces genomic instability and cancer; yeast carrying the analogous F391I allele shows classic minichromosome-loss

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— demonstrating deep conservation of the licensing/instability mechanism.

- **Transmission / zoonosis:** not applicable (genetic disease).

15. Model Organisms

- **Mouse (mammalian):**

- **Mcm4^{Chaos3} (F345I) hypomorph** (MGI): destabilizes MCM4:MCM6, fewer dormant origins, aphidicolin-hypersensitive fibroblasts, **>80% mammary adenocarcinoma in homozygous females** (~12-month latency); models genomic instability and cancer

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- **Mcm4-depleted / conditional adrenal studies:** reproduce **abnormal adrenal morphology** with non-steroidogenic GATA4⁺/Gli1⁺ cells replacing steroidogenic zona fasciculata — directly modeling the human adrenal phenotype

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- **Mcm4^{D573H} (Sdl)** dominant allele: T-cell lymphoblastic leukemia/lymphoma, chromosomal amplifications/deletions incl. *Notch1*

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- **Mcm4^{-/-} null:** preimplantation-lethal — establishes essentiality and why only hypomorphs are viable

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- **Epistasis models:** Mcm4^{Chaos3} × Atm/p21/Chk2 and × Fancc reveal DDR/FA-pathway modifiers

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- **Yeast (*S. cerevisiae*):** engineered F391I (ortholog of Chaos3) shows minichromosome-loss phenotype

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— tractable system for helicase-function assays.

- **In vitro / cellular:** patient-derived fibroblasts (genomic instability rescued by WT MCM4;

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primary NK-cell and activated-NK replication-stress assays (aphidicolin, CD56^{dim} apoptosis;

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- **Phenotype recapitulation:** mouse models capture **adrenal**, **genomic-instability**, and **cancer** features well; the **selective human NK CD56^{dim} defect** is best studied in human cells (mouse NK subsets differ). **Limitations:** rodents do not fully replicate the human CD56^{dim}/CD56^{bright} NK architecture or the exact growth phenotype.
- **Resources:** MGI (Mcm4 alleles), IMSR, SGD (MCM4/CDC54), Cellosaurus (patient fibroblasts).

Supported vs Refuted Hypotheses

Supported: - Biallelic hypomorphic MCM4 causes the disease (two independent human studies + functional rescue). - Selective CD56^{dim} NK loss arises from division-dependent maturation failure. - Adrenal insufficiency reflects steroidogenic-cell replacement by GATA4/Gli1⁺ cells. - Reduced dormant-origin licensing → replication stress → genomic instability → tissue-selective disease and cancer predisposition (cross-species).

Refuted / excluded: Not a complete MCM4 null (lethal); not a mineralocorticoid/aldosterone defect (glucocorticoid-selective); not a constitutional chromosomal rearrangement; not environmentally or infectious *caused* (infections are downstream triggers).

Limitations & Future Directions

- Very small human cohorts → no robust prevalence, survival, penetrance percentages, or quantified human cancer risk. Human cancer-risk inference rests on model organisms.

- No approved disease-specific therapy or clinical trials; HSCT experience in this exact genotype is minimal, and it cannot address non-hematopoietic features.
- Future needs: international patient registry / natural-history study; longitudinal cancer surveillance; genotype–phenotype expansion beyond the founder allele; evaluation of gene-correction and replication-stress-mitigating strategies.

Evidence-source key

Human

clinical/genetic:

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Model

organism

(mouse/yeast):

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